

Notes on Semigroups and Related Structures

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1 Elementary Concepts

1.1 Sets and Relations

1.1.1 Relations

Definition 1.1.1 (Binary relation). Let X be a set. Every subset ρ of the Cartesian product $X \times X$ is called a (*binary*) *relation* on X . If $(x, y) \in \rho$, we also write $x\rho y$, where $x, y \in X$.

Definition 1.1.2 (Domain and range). For a relation $\rho \subseteq X \times X$, the *domain* and *range* of R are defined by

$$\text{dom } \rho = \{x \in X : \exists y \in X \text{ with } (x, y) \in R\},$$

and

$$\text{ran } \rho = \{y \in X : \exists x \in X \text{ with } (x, y) \in R\}.$$

The word *image* is also used sometimes as a synonym of range.

Notation 1.1.3. The set of all binary relations on X is denoted by $\mathfrak{B}(X)$.

Remark 1.1.4. If $\rho, \tau \in \mathfrak{B}(X)$, then $\rho \subseteq \tau$ means that $x\rho y \Rightarrow x\tau y$ for all $x, y \in X$. We shall often write $\rho \leq \tau$ for $\rho \subseteq \tau$.

Definition 1.1.5 (Complement). The complement of a relation $\rho \in \mathfrak{B}(X)$ is the relation $\bar{\rho} \in \mathfrak{B}(X)$ defined by

$$(x, y) \in \bar{\rho} \iff (x, y) \notin \rho.$$

Definition 1.1.6 (Properties of a relation). A relation $\rho \in \mathfrak{B}(X)$ is called

1. *reflexive*, if $x\rho x$ for all $x \in X$;
2. *symmetric*, if $x\rho y$ implies $y\rho x$;
3. *antisymmetric*, if $x\rho y$ and $y\rho x$ together imply $x = y$;
4. *transitive*, if $x\rho y$ and $y\rho z$ imply $x\rho z$;
5. *right unique*, if ρy and $x\rho z$ imply $y = z$;
6. *left unique*, if $x\rho y$ and $z\rho y$ imply $x = z$;
7. *right total*, if for every $y \in X$ there exists $x \in X$ such that $x\rho y$;
8. *left total*, if for every $x \in X$ there exists $y \in X$ such that $x\rho y$.

The last four notions also apply to relations $\rho \subseteq X \times Y$ for two possibly different sets X and Y .

1.1.2 Mappings

Definition 1.1.7 (Mapping). Let X and Y be sets. A mapping $\varphi : X \rightarrow Y$ is a relation $\varphi \subseteq X \times Y$ which is right unique and left total; that is, for every $x \in X$ there exists exactly one $y \in Y$ such that $(x, y) \in \varphi$.

Remark 1.1.8. As usual, we write $\varphi(x) = y$. Sometimes we also write $x\varphi = y$. The set of all mappings from X to Y is denoted by $\text{Map}(X, Y)$ or Y^X . We also use $\mathcal{T}(X)$ for $\text{Map}(X, X)$. The elements of $\mathcal{T}(X)$ are called *transformations* of X .

Definition 1.1.9 (Injective, surjective, bijective). A mapping $\varphi : X \rightarrow Y$ is called

1. *injective*, if $\varphi(x) = \varphi(x') \Rightarrow x = x'$ for all $x, x' \in X$;
2. *surjective*, if for every $y \in Y$ there exists $x \in X$ such that $\varphi(x) = y$;
3. *bijective*, if it is both injective and surjective.

Equivalently, injectivity means that the relation is left unique, and surjectivity means that the relation is right total.

Remark 1.1.10. The subset of all bijective transformations in $\mathcal{T}(X)$ is denoted by $\mathcal{S}(X)$, and its elements are called *permutations* or *bijections* of X .

Definition 1.1.11 (Identity map). The identity map $\text{id}_X : X \rightarrow X$ is defined by

$$\text{id}_X(x) = x \quad \forall x \in X.$$

Definition 1.1.12 (Constant mapping). For $y \in Y$, the constant mapping $c_y : X \rightarrow Y$ is defined by

$$c_y(x) = y \quad \forall x \in X.$$

Definition 1.1.13 (Restriction). If $\varphi : X \rightarrow Y$ is a mapping and $A \subseteq X$, then the restriction of φ to A is defined by $\varphi|_A : A \rightarrow Y$ such that

$$\varphi|_A(a) = \varphi(a) \quad \forall a \in A.$$

Conversely, we say that φ extends $\varphi|_A$.

Definition 1.1.14 (Partial mapping). A partial mapping $\varphi : X \rightarrow Y$ is defined to be a right unique relation; that is, $\varphi : \text{dom } \varphi \rightarrow Y$ is a mapping, where $\text{dom } \varphi \subseteq X$. The set $\text{dom } \varphi$ is called the domain of φ . The set of all partial mappings on X (from X to X) is denoted by $\mathcal{PT}(X)$.

Remark 1.1.15. The mapping $\varphi : \text{dom } \varphi \rightarrow \text{im } \varphi$ with $\text{dom } \varphi = \text{im } \varphi = \emptyset$ is called the empty mapping.

1.1.3 Order Relations

Definition 1.1.16 (Partial order). A reflexive, antisymmetric, and transitive relation α on a set X is called a *partial ordering* on X , or simply an ordering of X . One writes $x \leq y$ for $(x, y) \in \alpha$, and the pair $(X, \leq)$ is called a *partially ordered set* or a *poset*.

Definition 1.1.17 (Total order). If in addition, a partial ordering has the property that for every $x, y \in X$, either $x \leq y$ or $y \leq x$. In this case, $\leq$ is called a *total order*. Then $(X, \leq)$ is also called a *linearly ordered set* or a *chain*.

Definition 1.1.18 (Maximal and minimal elements). Let X be a poset. An element $w \in X$ is called a *maximal* (respectively, *minimal*) element of $(X, \leq)$ if $w \leq u$ (respectively, $u \leq w$) implies $u = w$ for every $u \in X$.

Definition 1.1.19 (Minimal condition and well-ordering). A poset $(X, \leq)$ satisfies the *minimal condition* if every non-empty subset of X has a minimal element. In that case, we also say that $(X, \leq)$ is *well ordered*.

Definition 1.1.20 (Lower bound and meet). Let Y be a non-empty subset of a poset X . An element $c \in X$ is called a *lower bound* for Y if $c \leq y$ for all $y \in Y$. An element $u \in X$ is called the *greatest lower bound* or *meet* of Y if u is a lower bound for Y and $z \leq u$ for every lower bound z of Y . The greatest lower bound of Y is denoted by $\bigwedge y : y \in Y$. If $Y = \{a, b\}$, we also write $a \wedge b$.

Definition 1.1.21 (Lower semilattice). If $(X, \leq)$ is such that $a \wedge b$ exists for every pair $a, b \in X$, then $(X, \leq)$ is called a *lower semilattice*. If, in addition, $\bigwedge \{y : y \in Y\}$ exists for every non-empty subset $Y \subseteq X$, then $(X, \leq)$ is called a *complete lower semilattice*.

Definition 1.1.22 (Upper bound and join). Similarly, for a non-empty subset $Y \subseteq X$, an element $c \in X$ is called an *upper bound* for Y if $y \leq c$ for all $y \in Y$. An element $v \in X$ is called the *least upper bound* or *join* of Y if v is an upper bound for Y and $v \leq z$ for every upper bound z of Y . The least upper bound of Y is denoted by $\bigvee \{y : y \in Y\}$. If $Y = \{a, b\}$, we also write $a \vee b$.

Definition 1.1.23 (Upper semilattice). If $(X, \leq)$ is such that $a \vee b$ exists for every pair $a, b \in X$, then $(X, \leq)$ is called an *upper semilattice*. If, in addition, $\bigvee y : y \in Y$ exists for every non-empty subset $Y \subseteq X$, then $(X, \leq)$ is called a *complete upper semilattice*.

Definition 1.1.24 (Lattice). If a poset is both a (complete) lower semilattice and a (complete) upper semilattice then it is called a (complete) lattice. In this case we use the notation $(X, \leq, \wedge, \vee)$.

Remark 1.1.25. The set $\mathfrak{B}(X)$ of all binary relations on a set X is a poset under inclusion; that is, $\rho \leq \tau$ if and only if $\rho \subseteq \tau$, for $\rho, \tau \in \mathfrak{B}(X)$. Moreover, it is a lattice with

$$\rho \wedge \tau = \rho \cap \tau \quad \text{and} \quad \rho \vee \tau = \rho \cup \tau,$$

which is called the lattice of relations on X and is denoted by $(\mathfrak{B}(X), \leq, \wedge, \vee)$.

Lemma 1.1.1 (Zorn's lemma). *If $(X, \leq)$ is a non-empty poset such that every chain of elements has an upper bound in X , then X has at least one maximal element.*

Theorem 1.1.2 (Zermelo's theorem on well-ordering). *Every non-empty set can be well ordered.*

1.1.4 Equivalence relations and partitions

Definition 1.1.26. An *equivalence* or *equivalence relation* on X is a reflexive, symmetric and transitive relation on X .

Special equivalence relations on X are $\text{id}_X = \{(x, x) \mid x \in X\}$, the *diagonal relation* on X , also denoted by Δ_X , and the *total relation* $X \times X$, also denoted by ∇_X .

Definition 1.1.27. Let $K \subseteq X$ be a subset. Define the *Rees equivalence* ρ_K by setting

$$x \rho_K x' \iff x, x' \in K \text{ or } x = x'.$$

We denote the set of all equivalences on a set X by $\mathcal{E}(X)$. As a subset of $\mathfrak{B}(X)$ it is partially ordered by inclusion.

We call the intersection of all equivalences containing the relation $\rho \in \mathcal{B}(X)$ the *equivalence on X generated by ρ* or the *equivalence closure* of ρ , and denote it by ρ^e .

If $\{\rho_i \in \mathcal{E}(X) \mid i \in I\}$ is a non-empty set of equivalences on X , then $\bigcap_{i \in I} \rho_i$ is again an equivalence on X .

If $\rho \in \mathcal{B}(X)$, then the intersection of all equivalences containing ρ is the unique minimal equivalence on X containing ρ .

In fact $\mathcal{E}(X)$ is a lattice. If $\rho, \sigma \in \mathcal{E}(X)$, then their greatest lower bound $\rho \wedge \sigma$ is $\rho \cap \sigma$ and their least upper bound $\rho \vee \sigma$ is $(\rho \cup \sigma)^e$.

Definition 1.1.28. For $\rho \in \mathcal{E}(X)$ and $A \subseteq X$, set

$$\rho(A) := \{b \in X \mid \text{there exists } a \in A \text{ with } (a, b) \in \rho\}.$$

For $A = \{a\}$ we write instead of $\rho(a)$ also $[a]_\rho$ or just $[a]$, and call this set the ρ -class of a or the class of a (with respect to ρ).

Let $\rho, \sigma \in \mathcal{E}(X)$. Then $\rho \subseteq \sigma$ if and only if every σ -class is a union of ρ -classes.

Definition 1.1.29. A family of subsets X_i , $i \in I$, of a set X is called a *partition* of X if the subsets X_i are pairwise disjoint, non-empty, and their union is X .

If ρ is an equivalence relation on X , then the set of equivalence classes

$$\pi(\rho) = \{[x]_\rho \mid x \in X\}$$

is the partition of X corresponding to ρ .

On the other hand, if $\pi = \{X_i \mid i \in I\}$ is a partition of X into subsets X_i , then the relation

$$\rho(\pi) = \{(x, y) \in X \times X \mid \text{there exists } i \in I \text{ such that } x, y \in X_i\}$$

is an equivalence relation on X . Note that

$$\rho(\pi(\rho)) = \rho, \quad \pi(\rho(\pi)) = \pi.$$

The set of classes with respect to an equivalence relation ρ is called the *factor set* of X with respect to ρ and denoted by X/ρ . The mapping

$$\pi_\rho : X \rightarrow X/\rho, \quad x \mapsto [x]_\rho$$

is called the *canonical surjection* with respect to ρ .

For any $\rho \in \mathfrak{B}(X)$, let

$$\begin{aligned}\rho^R &= \bigcap \{\sigma \in B(X) \mid \rho \subseteq \sigma, \sigma \text{ is reflexive}\}, \\ \rho^S &= \bigcap \{\sigma \in B(X) \mid \rho \subseteq \sigma, \sigma \text{ is symmetric}\}.\end{aligned}$$

There is at least one element $\sigma \in \mathfrak{B}(X)$ satisfying the condition in each of the collections above, namely $\sigma = X \times X$. Furthermore, since every such σ contains ρ , the intersections ρ^R and ρ^S are both non-empty. It is easy to see that

- ρ^R , called the *reflexive closure* of ρ , is the smallest reflexive relation containing ρ ;
- ρ^S , called the *symmetric closure* of ρ , is the smallest symmetric relation containing ρ .

1.1.5 Inverse relation, restriction and composition

Let $\rho \subseteq X \times Y$. The *inverse* (*converse*) relation to ρ is

$$\rho^{-1} = \{(y, x) \in Y \times X \mid (x, y) \in \rho\}.$$

Similar as for mappings, we now define the *restriction* $\rho|_{X'}$ of a relation $\rho \in B(X)$ for a subset $X' \subseteq X$ by

$$\rho|_{X'} = \rho \cap (X' \times X').$$

The *composition* of relations $\rho, \sigma \in \mathfrak{B}(X)$ is defined as

$$\rho \circ_r \sigma = \{(x, z) \in X \times X \mid \text{there exists } y \in X \text{ with } x\sigma y \text{ and } y\rho z\},$$

if we start with the right relation σ , and analogously for $\rho \subseteq X \times Y, \sigma \subseteq Y \times Z$ as

$$\rho \circ_r \sigma = \{(x, z) \in X \times Z \mid \text{there exists } y \in Y \text{ with } x\sigma y \text{ and } y\rho z\}.$$

If we start with the left relation ρ , we define

$$\rho \circ_l \sigma = \{(x, z) \in X \times X \mid \text{there exists } y \in X \text{ with } x\rho y \text{ and } y\sigma z\},$$

for $\rho, \sigma \in B(X)$, and analogously for $\rho \subseteq X \times Y, \sigma \subseteq Y \times Z$ as

$$\rho \circ_l \sigma = \{(x, z) \in X \times Z \mid \text{there exists } y \in Y \text{ with } x\rho y \text{ and } y\sigma z\}.$$

Usually, for convenience, we will denote either $\circ_r$ or $\circ_l$ by $\circ$. We will be careful in any situation to indicate the use of either convention.

If $\varphi \in \text{Map}(X, Y)$ and $\psi \in \text{Map}(Y, Z)$, then in the case when mapping symbols are written to the left of the elements on which they act, by $\psi\varphi$ we mean $\psi \circ_r \varphi$, i.e.

$$(\psi\varphi)(x) = \psi(\varphi(x)) \quad \forall x \in X.$$

In the case when mapping symbols are written to the right of the elements on which they act, by $x\varphi\psi$ we mean $\varphi \circ_l \psi$, i.e.

$$(x\varphi)\psi = (x\varphi)\psi \quad \forall x \in X.$$

1.1.6 Transitive and equivalence closure

It is clear that the operation $\circ$ on $\mathfrak{B}(X)$ is associative and thus $(\mathfrak{B}(X), \circ)$ will be one of the examples of monoids (definition in a further section) with the identity element id_X . Sometimes we write $\rho\sigma$ for $\rho \circ \sigma$. We define powers of a relation $\rho \in \mathfrak{B}(X)$ as follows:

$$\rho^0 = id_X, \quad \rho^i = \rho \circ \rho \circ \dots \circ \rho, \text{ } i \text{ times, } i \in \mathbb{N}.$$

Definition 1.1.30. The *transitive closure* of a relation $\rho \in \mathfrak{B}(X)$ is the relation

$$\rho^\infty = \bigcup_{n=1}^{\infty} \rho^n \in \mathfrak{B}(X).$$

Lemma 1.1.3. If $\rho \in \mathfrak{B}(X)$, then ρ^∞ is the smallest transitive relation in $(\mathfrak{B}(X), \leq)$ containing ρ .

Proof. If $(x, y), (y, z) \in \rho^\infty$, then there exist $m, n \in \mathbb{N}$ such that $(x, y) \in \rho^m, (y, z) \in \rho^n$, and

$$(x, z) \in \rho^m \circ \rho^n = \rho^{m+n} \subseteq \rho^\infty.$$

Thus ρ^∞ is a transitive relation and $\rho \subseteq \rho^\infty$. If now σ is a transitive relation on X and $\rho \subseteq \sigma$, then

$$\rho^2 = \rho \circ \rho \subseteq \sigma \circ \sigma \subseteq \sigma.$$

By induction it follows easily that $\rho^n \subseteq \sigma$ for all $n \in \mathbb{N}$, i.e. $\rho^\infty \subseteq \sigma$. □

Theorem 1.1.4. If $\rho \in \mathfrak{B}(X)$, then for the equivalence closure ρ^e we get

$$\rho^e = (\rho \cup \rho^{-1} \cup id_X)^\infty.$$

Proof. It is clear that $\tau = (\rho \cup \rho^{-1} \cup id_X)^\infty$ contains ρ and is transitive and reflexive. The relation $\nu = \rho \cup \rho^{-1} \cup id_X$ is symmetric and therefore $\nu^n = (\nu^{-1})^n = (\nu^n)^{-1}$. Thus, if $(x, y) \in \tau$, then $(x, y) \in \nu^n$ for some $n \in \mathbb{N}$ and therefore $(y, x) \in \nu^n$. Thus $(y, x) \in \tau$, i.e. τ is symmetric. This means that $\tau \in \mathcal{E}(X)$.

If $\sigma \in \mathcal{E}(X)$ and $\rho \subseteq \sigma$, then $id_X \subseteq \sigma, \rho^{-1} \subseteq \sigma^{-1} = \sigma$, and therefore $\nu = \rho \cup \rho^{-1} \cup id_X \subseteq \sigma$. Since $\nu \circ \nu \subseteq \sigma \circ \sigma \subseteq \sigma$, we get $\nu^n \subseteq \sigma$ for all $n \in \mathbb{N}$, and hence $\tau \subseteq \sigma$. □

Corollary 1.1.5. If $\rho \in \mathfrak{B}(X)$, then $(x, y) \in \rho^e$ if and only if $x = y$ or for some $n \in \mathbb{N}$ there exists a sequence of elements $x_0, x_1, x_2, \dots, x_n$ in X such that $x_0 = x, x_n = y$, and for every $i \in \{1, \dots, n\}$ either $(x_{i-1}, x_i) \in \rho$ or $(x_i, x_{i-1}) \in \rho$.

In particular, if ρ and σ are equivalence relations on a set X , then in $\mathcal{E}(X)$ their join $\rho \vee \sigma$ is the relation defined by

$$x(\rho \vee \sigma)y \iff \text{there exist } x_0, x_1, \dots, x_n \in X \text{ such that } x = x_0, y = x_n$$

and

$$(x_{i-1}, x_i) \in \tau_i, \quad \tau_i \in \{\rho, \sigma\}, \quad i \in \{1, \dots, n\}.$$

Proposition 1.1.6. For any $\rho \in \mathfrak{B}(X)$,

1. $\rho^R = \rho \cup id_X$;
2. $\rho^S = \rho \cup \rho^{-1}$;
3. $(\rho^R)^S = (\rho^S)^R = \rho \cup \rho^{-1} \cup id_X$;
4. $\rho^e = \left((\rho^R)^S \right)^\infty$.

1.1.7 Kernel equivalence and Homomorphism Theorem for sets

Definition 1.1.31. Let $\varphi : X \rightarrow Y$ be a mapping. The relation

$$\ker \varphi = \{(x, x') \in X \times X \mid \varphi(x) = \varphi(x')\}$$

is an equivalence on X which is called the *kernel equivalence* or the *kernel* of φ . It follows from the definition of the composition of relations that

$$\ker \varphi = \varphi^{-1} \circ \varphi = \varphi^{-1} \circ_r \varphi.$$

Theorem 1.1.7 (Homomorphism Theorem for sets). Let $\varphi : X \rightarrow Y$ be a mapping and let ρ be an equivalence on X such that $x\rho x'$ implies $\varphi(x) = \varphi(x')$, i.e. $\rho \leq \ker \varphi$. Then $\varphi' : X/\rho \rightarrow Y$, with $\varphi'([x]_\rho) := \varphi(x)$, $x \in X$, is the unique mapping such that the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & Y \\ \pi_\rho \downarrow & \nearrow \varphi' & \\ X/\rho & & \end{array}$$

i.e. such that $\varphi' \pi_\rho = \varphi$, where π_ρ denotes the canonical surjection.

If $\rho = \ker \varphi$, then φ' is injective, and if φ is surjective, then φ' is surjective.

Proof. Define $\varphi' : X/\rho \rightarrow Y$ by $\varphi'([x]_\rho) = \varphi(x)$ for any $x \in X$. Suppose $[x]_\rho = [x']_\rho$ for $x, x' \in X$. Then $(x, x') \in \rho$ and thus $\varphi(x) = \varphi(x')$. Hence φ' is well-defined.

Suppose $\varphi'([x]_\rho) = \varphi'([x']_\rho)$. Then $\varphi(x) = \varphi(x')$ and thus $(x, x') \in \ker \varphi$.

If $\rho = \ker \varphi$, then $[x]_\rho = [x']_\rho$ and thus φ' is injective. □

1.1.8 Congruences

Definition 1.1.32. A relation ρ on X is called

- *left compatible* if for all $x, y, z \in X$,

$$(x, y) \in \rho \implies (zx, zy) \in \rho;$$

- *right compatible* if for all $x, y, z \in X$,

$$(x, y) \in \rho \implies (xz, yz) \in \rho;$$

- *compatible* if for all $x, y, s, t \in X$,

$$(x, y) \in \rho \text{ and } (s, t) \in \rho \implies (xs, yt) \in \rho.$$

A left compatible equivalence relation is called a *left congruence*; a right compatible equivalence relation is called a *right congruence*; and a compatible equivalence relation is called a *congruence*.

Proposition 1.1.8. A relation ρ on X is a congruence if and only if it is both a left and a right congruence.

1 Elementary Concepts

Proof. Suppose ρ is both a left and a right congruence. Let $x, y, z, t \in X$ be such that $(x, y) \in \rho$ and $(z, t) \in \rho$. Since ρ is a right congruence, $(xz, yt) \in \rho$, and since ρ is a left congruence, $(yz, yt) \in \rho$. By transitivity of ρ , it follows that $(xz, yt) \in \rho$. Hence ρ is a congruence.

Conversely, suppose ρ is a congruence. Let $x, y \in X$ be such that $(x, y) \in \rho$, and let $z \in X$. Since ρ is reflexive, $(z, z) \in \rho$. By compatibility, $(zx, zy) \in \rho$. Thus ρ is left compatible. Similarly, $(xz, yz) \in \rho$, so ρ is right compatible. Hence ρ is both a left and a right congruence. $\square$

For any $\rho \in \mathfrak{B}(X)$, let

$$\rho^c = \bigcap \{ \sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is left and right compatible} \}.$$

Then ρ^c is the smallest left and right compatible relation containing ρ . Also, let

$$\rho^\# = \bigcap \{ \sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is a congruence} \}.$$

Then ρ^c , called the congruence generated by ρ , is the smallest congruence containing ρ .

Proposition 1.1.9. For any $\rho, \sigma \in \mathfrak{B}(X)$,

1. $(\rho \cup \sigma)^c = \rho^c \cup \sigma^c$;
2. $(\rho^{-1})^c = (\rho^c)^{-1}$.

Proposition 1.1.10. For any $\rho \in \mathfrak{B}(X)$, we have

$$\rho^\# = (\rho^c)^e.$$

Proof. By definition, $(\rho^c)^e$ is an equivalence relation containing ρ^c , and hence containing ρ . Notice that

$$\begin{aligned} (\rho^c)^e &= (\rho^c \cup (\rho^c)^{-1} \cup id_X)^\infty \\ &= ((\rho \cup \rho^{-1} \cup id_X)^c)^\infty \\ &= \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \end{aligned}$$

Now, for all $x, y, z \in X$,

$$\begin{aligned} &(x, y) \in (\rho^c)^e \\ \implies &(x, y) \in \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &\exists n \in \mathbb{N} \text{ such that } (x, y) \in ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &\exists x_0, x_1, \dots, x_n \text{ such that } x = x_0 \text{ and } y = x_n \text{ and } (\forall i) (x_i, x_{i+1}) \in (\rho \cup \rho^{-1} \cup id_X)^c \\ \implies &zx = zx_0, zx_n = zy \text{ and } (\forall i) (zx_i, zx_{i+1}) \in (\rho \cup \rho^{-1} \cup id_X)^c \\ &\quad (\text{since } (\rho \cup \rho^{-1} \cup id_X)^c \text{ is left and right congruence}) \\ \implies &(zx, zy) \in ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &(zx, zy) \in \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &(zx, zy) \in (\rho^c)^e. \end{aligned}$$

Therefore, $(\rho^c)^e$ is left compatible. Similarly, $(\rho^c)^e$ is right compatible. Hence $(\rho^c)^e$ is a congruence containing ρ .

Finally, let τ be any congruence containing ρ . Then τ is left and right compatible, so $\rho^c \subseteq \tau$. Since τ is an equivalence relation, it must contain $(\rho^c)^e$. Hence $(\rho^c)^e \subseteq \tau$. Therefore, $(\rho^c)^e$ is the smallest congruence containing ρ , and therefore $\rho^\# = (\rho^c)^e$. Consequently, $\rho^\# = (\rho^c)^e$ $\square$

We denote by $\mathcal{C}(X)$ the set of all congruences on X . Then $\mathcal{C}(X)$, as a subset of $\mathcal{E}(X)$ and $\mathcal{B}(X)$, is partially ordered by inclusion.

In fact, $\mathcal{C}(X)$ forms a lattice. For $\rho, \sigma \in \mathcal{C}(X)$, we have

$$\rho \wedge \sigma = \rho \cap \sigma \quad \text{and} \quad \rho \vee \sigma = (\rho \cup \sigma)^\#.$$

Also,

$$\begin{aligned} (\rho \cup \sigma)^\# &= ((\rho \cup \sigma)^c)^e \\ &= (\rho^c \cup \sigma^c)^e \\ &= (\rho \cup \sigma)^e \end{aligned}$$

so there is no ambiguity in writing $\rho \vee \sigma$ whether one means the join $(\rho \vee \sigma)^e$ in the lattice $\mathcal{E}(X)$ or the join $(\rho \vee \sigma)^\#$ in the lattice $\mathcal{C}(X)$.

Proposition 1.1.11. *Let $\rho, \sigma \in \mathcal{C}(X)$. Then*

$$\rho \vee \sigma = (\rho \circ \sigma)^\infty.$$

Proof. Let $\rho, \sigma \in \mathcal{C}(X)$. Since $\rho \vee \sigma$ contains both ρ and σ , it follows that

$$\rho \circ \sigma \subseteq (\rho \cup \sigma) \circ (\rho \cup \sigma) = (\rho \vee \sigma)^2.$$

More generally, $(\rho \circ \sigma)^n \subseteq (\rho \cup \sigma)^{2n}$. Thus,

$$(\rho \circ \sigma)^\infty = \bigcup_{n=0}^{\infty} (\rho \circ \sigma)^n \subseteq \bigcup_{n=0}^{\infty} (\rho \cup \sigma)^{2n} = (\rho \cup \sigma)^\infty.$$

On the other hand, $\rho \circ \sigma \supseteq \rho \circ \text{id}_X = \rho$ (since σ is reflexive), and $\rho \circ \sigma \supseteq \text{id}_X \circ \sigma = \sigma$ (since ρ is reflexive). Thus, $\rho \cup \sigma \subseteq \rho \circ \sigma$. Hence, $(\rho \cup \sigma)^\infty \subseteq (\rho \circ \sigma)^\infty$.

Consequently, $(\rho \cup \sigma)^\infty = (\rho \circ \sigma)^\infty$.

Then,

$$\begin{aligned} \rho \vee \sigma &= (\rho \cup \sigma)^e = \left(((\rho \cup \sigma)^R)^S \right)^\infty \\ &= ((\rho \cup \sigma) \cup (\rho \cup \sigma)^{-1} \cup \text{id}_X)^\infty \\ &= (\rho \cup \sigma \cup \rho^{-1} \cup \sigma^{-1} \cup \text{id}_X)^\infty \\ &= (\rho \cup \sigma \cup \text{id}_X)^\infty \\ &= (\rho \cup \sigma)^\infty = (\rho \circ \sigma)^\infty. \end{aligned}$$

This completes the proof. □

Proposition 1.1.12. *Let $\rho, \sigma \in \mathcal{C}(X)$. If $\rho \circ \sigma = \sigma \circ \rho$, then*

$$\rho \vee \sigma = \rho \circ \sigma.$$

Proof. Suppose $\rho \circ \sigma = \sigma \circ \rho$. Then

$$(\rho \circ \sigma)^2 = \rho \circ \sigma \circ \rho \circ \sigma = \rho \circ \rho \circ \sigma \circ \sigma = \rho \circ \sigma,$$

since ρ and σ are transitive.

Hence, $(\rho \circ \sigma)^n = \rho \circ \sigma$ for all $n \in \mathbb{N}$, and therefore

$$\rho \circ \sigma = (\rho \circ \sigma)^\infty = \bigcup_{n \geq 0} (\rho \circ \sigma)^n = \rho \circ \sigma.$$

So $\rho \vee \sigma = \rho \circ \sigma$. □

1.2 Introductory Algebraic structures

1.2.1 Groupoids, Identities, and Zeros

Definition 1.2.1. A *groupoid* is a non-empty set A together with a binary operation

$$A \times A \rightarrow A, \quad (a, a') \mapsto aa'.$$

The operation is usually called the *multiplication* of the groupoid.

Remark 1.2.2. Let A be a groupoid. For non-empty subsets $B, C \subseteq A$, define

$$BC = \{bc \mid b \in B, c \in C\}.$$

We write $B^2 = BB$, and similarly for higher products.

Definition 1.2.3. Let S be a groupoid. An element $e \in S$ is called

- a *left identity* if $es = s$ for all $s \in S$;
- a *right identity* if $se = s$ for all $s \in S$;
- an *identity* if $es = s = se$ for all $s \in S$.

An identity is also called a *neutral element*.

Proposition 1.2.1. *If a groupoid S has both a left and a right identity, then they coincide. Hence a groupoid has at most one identity.*

Proof. Let e' be a left identity and e'' a right identity. Then

$$e'' = e'e'' = e'.$$

□

We denote the identity (when it exists) by 1 or more precisely by 1_S .

Definition 1.2.4. Let S be a groupoid with identity.

- $s \in S$ is *left invertible* if there exists $t \in S$ such that $ts = 1$.
- s is *right invertible* if there exists $t \in S$ such that $st = 1$.
- s is *invertible* if there exists $t \in S$ satisfying $ts = st = 1$.

In third case, such an element t is called an inverse of s , and is denoted by s^{-1} whenever it is unique.

Definition 1.2.5. Let S be a groupoid. An element $z \in S$ is called

- a *left zero* if $zs = z$ for every $s \in S$;
- a *right zero* if $sz = z$ for every $s \in S$;
- a *zero* if $zs = z = sz$ for every $s \in S$.

A zero is usually denoted by 0.

Exercise 1.2.2. *Show that if a groupoid has both a left zero and a right zero, then they coincide. Hence a groupoid has at most one zero.*

1.2.2 Semigroups, monoids and groups

Definition 1.2.6. A *semigroup* is a groupoid whose multiplication is associative. A semigroup with an identity element is called a *monoid*.

Proposition 1.2.3. *If an element of a monoid has both a left and a right inverse, then they coincide.*

Proof. Suppose $ts = 1$ and $st' = 1$. Then

$$t' = (ts)t' = t(st') = t. \quad \square$$

Corollary 1.2.4. *Every element of a monoid has at most one inverse.*

Definition 1.2.7. A *group* G is a monoid in which every element is invertible.

For any $a, b \in G$,

$$(ab)^{-1} = b^{-1}a^{-1}, \quad (a^{-1})^{-1} = a.$$

Exercise 1.2.5. *Suppose S is a semigroup with an element e satisfying $es = s$ for every $s \in S$, and suppose every $s \in S$ has an element $t \in S$ such that $ts = e$. Show that S is a group.*

Remark 1.2.8 (Free semigroup and free monoid). Let X be a non-empty set, called an *alphabet*. Define

$$X^+ = \{x_1x_2 \cdots x_n \mid n \in \mathbb{N}, x_i \in X, 1 \leq i \leq n\},$$

the set of all finite non-empty words over X .

Two words

$$x_1x_2 \cdots x_n, \quad y_1y_2 \cdots y_m$$

are equal if and only if

$$n = m \quad \text{and} \quad x_i = y_i \quad \text{for all } 1 \leq i \leq n.$$

Define the multiplication (concatenation) of words by

$$(x_1x_2 \cdots x_n)(y_1y_2 \cdots y_m) = x_1x_2 \cdots x_ny_1y_2 \cdots y_m.$$

With this operation, X^+ forms a semigroup, called the *free semigroup* on X , or the *free semigroup with basis X* .

Now let

$$X^* = X^+ \cup \{\varepsilon\},$$

where ε denotes the empty word. Then X^* is a monoid under concatenation, with identity element ε . It is called the *free monoid* on X , or the *free monoid with basis X* .

1.2.3 Examples

Example 1.2.9 (Groupoids). 1. $(\mathbb{Z}, -)$.

2. $(\mathbb{Q} \setminus \{0\}, /)$.

3. $(\mathbb{N}, \uparrow)$, where $n \uparrow m = n^m$.

4. The vector product $(\mathbb{R}^3, \times)$.

1 Elementary Concepts

Example 1.2.10 (Semigroups and monoids). 1. If $(R, +, \cdot)$ is a ring, then $(R, +)$ is a group and $(R, \cdot)$ is a semigroup.

2. $(\mathbb{N}, \cdot)$ is a monoid.

3. $(\mathbb{N}, +)$ is a semigroup.

4. $(\mathcal{T}(X), \circ)$, the full transformation monoid.

5. $(\mathcal{PT}(X), \circ)$, the monoid of partial transformations.

6. $(\mathfrak{B}(X), \circ)$, the semigroup of binary relations.

7. The free semigroup X^+ .

8. The free monoid X^* .

9. $(\mathbb{Z}, \gcd)$.

10. **The bicyclic monoid:**

Let $B = \mathbb{N}_0 \times \mathbb{N}_0$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Define a binary operation on B by

$$(m, n)(p, q) = (m - n + \max\{n, p\}, q - p + \max\{n, p\}).$$

Equivalently,

$$(m, n)(p, q) = \begin{cases} (m - n + p, q), & \text{if } p \geq n, \\ (m, n - p + q), & \text{if } p \leq n. \end{cases}$$

Then $(B, \cdot)$ is a monoid, called the *bicyclic monoid*, with identity element $(0, 0)$.

Example 1.2.11 (Groups). 1. $(\mathbb{Z}, +)$.

2. $(\mathbb{Q}, +)$.

3. $(\mathbb{Q} \setminus \{0\}, \cdot)$.

4. The residue class group $(\mathbb{Z}_r, +)$.

5. The symmetric group $S(X)$, or S_n when $|X| = n$.

2 Semigroups

2.1 Basic concepts and examples

Throughout this chapter, S will denote a semigroup with operation $\circ$. Formally, we write $(S, \circ)$ to indicate that we are considering the set S with the operation $\circ$, but we will only do this when we need to distinguish a particular operation. Unless we need to distinguish between different operations, we will often write xy instead of $x \circ y$ (where $x, y \in S$), and we will call the operation multiplication and the element xy (i.e. the result of applying the operation to x and y) the product of x and y .

Since the operation is associative, there is no ambiguity in the product $s_1 s_2 \cdots s_n$ (where each $s_i \in S$): the product is the same regardless of how we insert the brackets.

Any group is also a semigroup. The most familiar example of a semigroup that is not a group is the set of natural numbers $\mathbb{N}$ under the operation of addition. This is not a group since it contains no identity. (For our purposes, $\mathbb{N} = \{1, 2, 3, \dots\}$ does not include 0.)

Example 2.1.1. The trivial semigroup contains only one element e , with multiplication obviously defined by $ee = e$. Since e is (trivially) an identity, this semigroup is also called the trivial monoid.

Definition 2.1.2. A null semigroup is a semigroup with a zero z in which the product of any two elements is z . It is easy to see that this multiplication is associative. Notice that we can define a null semigroup on any set by choosing some element z and defining all products to be z .

Definition 2.1.3. If every element of S is a left zero (that is, $xy = x$ for all $x, y \in S$), then S is a left zero semigroup. If every element of S is a right zero (that is, $xy = y$ for all $x, y \in S$), then S is a right zero semigroup. We can define a left zero semigroup on any set X by defining the multiplication $xy = x$ for all $x, y \in X$; it is easy to see that this multiplication is associative.

If a semigroup does not contain identity (zero element), we can always add a new element to act as an identity (zero element). To see this, let 1 be a new element not in the semigroup S . Extend the multiplication on S to $S \cup \{1\}$ by $1x = x1 = x$ for all $x \in S$ and $11 = 1$. It is easy to prove that this extended multiplication is associative. Then $S \cup \{1\}$ is a monoid with identity 1 . Similarly, let 0 be a new element not in S and extend the multiplication on S to $S \cup \{0\}$ by $0x = x0 = 00 = 0$ for all $x \in S$. Again, this extended multiplication is associative. Then $S \cup \{0\}$ is a semigroup with zero 0 .

For any semigroup S , define

$$S^1 = \begin{cases} S, & \text{if } S \text{ has an identity,} \\ S \cup \{1\}, & \text{otherwise,} \end{cases} \quad S^0 = \begin{cases} S, & \text{if } S \text{ has a zero,} \\ S \cup \{0\}, & \text{otherwise.} \end{cases}$$

The semigroups S^1 and S^0 are called, respectively, the monoid obtained by adjoining an identity to S if necessary and the semigroup obtained by adjoining a zero to S if necessary.

2 Semigroups

Throughout this chapter, mappings are written on the right and composed left to right. To clarify: let $\varphi : X \rightarrow Y$ and $\psi : Y \rightarrow Z$ be maps. The result of applying φ to an element x of X is denoted $x\varphi$. The composition of φ and ψ is denoted $\varphi\psi$ or simply $\varphi\psi$, and is a map from X to Z with $x(\varphi\psi) = (x\varphi)\psi$ for all $x \in X$.

Definition 2.1.4 (Cartesian Product). Let $X = \{X_i : i \in I\}$ be a collection of sets. Informally, the Cartesian product $\prod_{i \in I} X_i$ of the sets in X is the set of tuples with $|I|$ components, where, for each $i \in I$, the i -th component is an element of X_i . More formally, the Cartesian product $\prod_{i \in I} X_i$ is the set of mappings σ from I to $\bigcup_{i \in I} X_i$ such that $i\sigma \in X_i$ for each $i \in I$.

We think the map σ as a tuple with i -th component $i\sigma$. We will use both mapping notation and (especially when the index set I is finite) tuple notation for elements of Cartesian products. When $I = \{1, \dots, n\}$ is finite, we write $X_1 \times \dots \times X_n$ for $\prod_{i \in I} X_i$. When the sets X_i are all equal to a set X , that is, when we consider products of $|I|$ copies of the set X , we write X^I for $\prod_{i \in I} X_i$.

Definition 2.1.5. Let $S = \{S_i : i \in I\}$ be a collection of semigroups. The direct product of the semigroups in S is their Cartesian product $\prod_{i \in I} S_i$ with componentwise multiplication: $(i)(s)(i)(t) = (i)(s)(i)t$, or, using tuple notation, if $(\dots, s_i, \dots), (\dots, t_i, \dots) \in \prod_{i \in I} S_i$, then

$$(\dots, s_i, \dots)(\dots, t_i, \dots) = (\dots, s_i t_i, \dots).$$

It is easy to prove that this componentwise multiplication is associative, and so the direct product is itself a semigroup.

For $x \in S$ and $n \in \mathbb{N}$, define

$$x^n = \underbrace{x \cdots x}_{n \text{ times}}.$$

In general, x^n is only defined for positive n . If S is a monoid, define $x^0 = 1_S$. An element x of S is *idempotent* if $x^2 = x$. The set of idempotents of S is denoted $E(S)$.

For any subsets X and Y of S , define $XY = \{xy : x \in X, y \in Y\}$. Write xY for $\{x\}Y$ and XY for $X\{y\}$. Since multiplication in S is associative, so is this product of subsets: for subsets X, Y, Z of S , we have $X(YZ) = (XY)Z$. By analogy with the above, for $X \subseteq S$ and $n \in \mathbb{N}$, define

$$X^n = \underbrace{X \cdots X}_{n \text{ times}}.$$

Definition 2.1.6 (Nilpotent Semigroup). The semigroup S is nilpotent if it contains a zero and there exists some $n \in \mathbb{N}$ such that $S^n = \{0\}$.

Definition 2.1.7. The semigroup S is a nilsemigroup if it contains a zero and for every $x \in S$, there exists some $n \in \mathbb{N}$ such that $x^n = \{0\}$.

Definition 2.1.8 (Cancellative Semigroup). The semigroup S is left cancellative if

$$(\forall x, y, z \in S)(zx = zy \Rightarrow x = y);$$

right cancellative if

$$(\forall x, y, z \in S)(xz = yz \Rightarrow x = y);$$

and cancellative if it is both left and right cancellative.

Note 2.1.9. A non-trivial semigroup with zero cannot be cancellative.

Definition 2.1.10 (Commutativity). The semigroup S is commutative if $xy = yx$ for all $x, y \in S$. For instance, $(\mathbb{Z}, +)$ and $(\mathbb{Z}, \cdot)$ are both commutative.

A non-trivial left zero semigroup is not commutative.

Let $x \in S$. If there is an element $y \in S$ such that $xyx = x$, then the element x is *regular*. Notice that in this case, $(xy)^2 = xyxy = x(yx)y = xy$ and $(yx)^2 = yxyx = y(xy)x = yx$ and hence xy and yx are idempotent elements of S .

Definition 2.1.11 (Regular Semigroup). If every element of S is regular, then S is a regular semigroup.

Now we define the inverse of an element of a semigroup S in general.

Definition 2.1.12. Let $x \in S$. An element $x' \in S$ is called as inverse of x if $x = xx'x$ and $x' = x'xx'$.

Notice that this is entirely different from the notion of left/right inverses above. We will never use *inverse* (on its own) to refer to a left or right inverse.

Proposition 2.1.1 (). Let $x \in S$. Then x has an inverse if and only if x is regular.

Proof. Obviously if x has an inverse, then it is regular. So suppose x is regular. Then there exists $y \in S$ such that $xyx = x$. Let $x' = yxy$. Then

$$xx'x = x(yxy)x = (xyx)yxyx = xyx = x$$

and

$$x'xx' = (yxy)x(yxy) = y(xy x)yxy = yxy = x',$$

so x' is an inverse of x . □

In the proof of Proposition 2.1.1, the element y might not be an inverse of x ; for example, let S be a semigroup with a zero and let $x = 0$ and $y \neq 0$.

An element x can have more than one inverse. The set of inverses of x is denoted $V(x)$. Notice also that a zero of a semigroup has an inverse, namely itself. In general, if $e \in S$ is idempotent, then $e^2 = e$ and so e is an inverse of itself. In particular, every idempotent is regular.

Example 2.1.13. (a) Let $U = \{0, \dots, k\}$ for some $k > 0$. Define $m \Delta n = \min\{m, n\}$. It is easy to see that Δ is associative, and so (U, Δ) is a semigroup. Notice that $0 \Delta m = m \Delta 0 = 0$ and $k \Delta m = m \Delta k = m$ for all $m \in U$. Hence U has zero 0 and identity k . Furthermore, $m \Delta m = m$ for all $m \in U$, so every element of U is idempotent. Finally, $m \Delta n = n \Delta m$ for all $m, n \in U$, so U is commutative.

(b) Define a similar associative operation Δ on $\mathbb{N}_0 \cup \{0\} = \{0, 1, 2, \dots\}$ by

$$m \Delta n = \min\{m, n\}.$$

Then $(\mathbb{N}, \Delta)$ has a zero 0 but has no identity. It is commutative and all its elements are idempotents.

2 Semigroups

(c) Consider the set of all 2×2 integer matrices:

$$M_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z} \right\}.$$

With the usual matrix multiplication, $M_2(\mathbb{Z})$ is a monoid with identity

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and zero

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

It is easy to see that $M_2(\mathbb{Z})$ is not commutative, that not all of its elements are idempotents. Since $M_2(\mathbb{Z})$ contains a zero, it is not cancellative.

(d) Now let V be the set of all 2×2 integer matrices with non-zero determinant. Again, V is a monoid. Let $P, Q, R \in V$. Suppose $RP = RQ$. Since $\det R \neq 0$, the matrix R has an inverse $R^{-1} \in M_2(\mathbb{Q})$. Hence V is left-cancellative. Similarly, it is right-cancellative and therefore cancellative.

Definition 2.1.14. Let L be a left zero semigroup and R a right zero semigroup. Let $B = L \times R$. This semigroup is called as $|L| \times |R|$ rectangular band, or simply a *rectangular band*.

For $(\ell_1, r_1), (\ell_2, r_2) \in B$, we have

$$(\ell_1, r_1)(\ell_2, r_2) = (\ell_1, r_2),$$

since (in particular) ℓ_1 is a left zero and r_2 is a right zero. Thus every element of B is idempotent, since $(\ell, r)(\ell, r) = (\ell, r)$ for all $(\ell, r) \in B$. Furthermore, for any $(\ell_1, r_1), (\ell_2, r_2) \in B$, we have

$$(\ell_1, r_1)(\ell_2, r_2)(\ell_1, r_1) = (\ell_1 \ell_2 \ell_1, r_1 r_2 r_1) = (\ell_1, r_1),$$

$$(\ell_2, r_2)(\ell_1, r_1)(\ell_2, r_2) = (\ell_2 \ell_1 \ell_2, r_2 r_1 r_2) = (\ell_2, r_2).$$

Hence (ℓ_2, r_2) is an inverse of (ℓ_1, r_1) . Thus every element is an inverse of every element. The name *rectangular band* comes from the following geometric interpretation of the multiplication. To find the product $(\ell_1, r_1)(\ell_2, r_2)$, take the rectangle two of whose vertices have coordinates (ℓ_1, r_1) and (ℓ_2, r_2) . Then the product is the vertex vertically in line with (ℓ_1, r_1) and horizontally in line with (ℓ_2, r_2) , as shown in Figure 2.1.

